Tech Bite #8: Inside the Radio -- The Transceiver

What the parts inside a radio are called and what each one does -- receiver, transmitter, oscillator, mixer, and modulation.

Group T7A · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

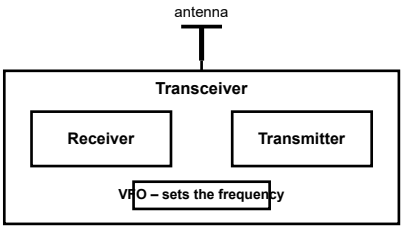
The Big Picture

A **transceiver** is a receiver and a transmitter in one box. This Bite names the parts inside and what each does – the vocabulary the exam uses for the building blocks of a radio. You will not repair these circuits; you just need to know the words.

Receiver and Transmitter

A **transceiver** is simply a **receiver and a transmitter combined** in one unit. (T7A02) Two words describe how well a receiver works:

- **Sensitivity** is its ability to **detect a weak signal**. (T7A01)
- **Selectivity** is its ability to **separate one signal** from others nearby. (T7A04)



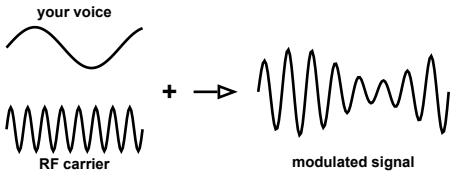
A transceiver holds both a receiver and a transmitter; the VFO sets the frequency for both.

The Building Blocks

- An **oscillator** generates a signal at a specific frequency. (T7A05) The **VFO** – Variable Frequency Oscillator – is the one that **sets your transmit and receive frequency**. (T7A11)
- A **mixer** converts a signal from one frequency to another. (T7A03)
- A **transverter** converts a transceiver's input and output to **another band**. (T7A06)
- The **PTT** (push-to-talk) input switches the transceiver **from receive to transmit** when it is grounded. (T7A07)
- An **RF power amplifier** added to the output **increases transmitted power**. (T7A10) On some VHF amplifiers, an SSB / CW-FM switch sets the amplifier for **proper operation in the chosen mode**. (T7A09)

Modulation

Putting your voice onto a radio signal has a name. **Modulation** is **combining speech with an RF carrier**. (T7A08) The carrier is the steady radio wave; modulation is what makes it carry your voice across the miles.



Modulation: your voice is combined with a steady RF carrier to make the signal that is transmitted.

Mental Cheat Sheet

- A **transceiver** = a receiver + a transmitter in one box.
- **Sensitivity** = ability to detect a weak signal.
- **Selectivity** = ability to separate nearby signals.
- An **oscillator** generates a signal; the **VFO** sets your frequency.
- A **mixer** converts a signal to a different frequency.
- A **transverter** moves a radio to another band.
- **PTT** switches receive-to-transmit when grounded.
- An **RF power amplifier** boosts transmitted power.
- **Modulation** = combining speech with an RF carrier.

Common Traps

- **Sensitivity** detects weak signals; **selectivity** separates nearby ones – do not swap them.
- The **VFO** sets the frequency; it is not automatic frequency control.
- A **mixer** changes frequency; an amplifier changes power.
- PTT switches to transmit when **grounded**.
- To increase transmit power, add an **RF power amplifier** – not a potentiometer.
- **Modulation** is voice on a carrier – not impedance matching or filtering.

Takeaway: A **transceiver** is a receiver and transmitter together. The receiver is judged by **sensitivity** (detecting weak signals) and **selectivity** (separating nearby ones). Inside, an **oscillator** (the VFO) sets the frequency, a **mixer** shifts frequencies, and **modulation** puts your voice on the carrier. An **RF power amplifier** boosts the output. Learn the words – meeting 8′s Dessert covers what to do when something goes wrong.

Bite 8 Question Bank (T7A)

T7A01: Which term describes the ability of a receiver to detect the presence of a signal?
A. RF gain
B. Selectivity
C. Sensitivity
D. Harmonic Distortion

T7A02. What is a transceiver?
A. A device that combines a receiver and transmitter
B. A device for matching feed line impedance to 50 ohms
C. A device for automatically sending and decoding Morse code
D. A device for converting receiver and transmitter frequencies to another band

T7A03. Which of the following is used to convert a signal from one frequency to another?
A. Phase splitter
B. Mixer
C. Inverter
D. Amplifier

T7A04. Which term describes the ability of a receiver to discriminate between multiple signals?
A. Discrimination ratio
B. Sensitivity
C. Selectivity
D. Harmonic distortion

T7A05. What is the name of a circuit that generates a signal at a specific frequency?
A. Reactance modulator
B. Phase modulator
C. Low-pass filter
D. Oscillator

T7A06. What device converts the RF input and output of a transceiver to another band?
A. High-pass filter
B. Low-pass filter
C. Transverter
D. Phase converter

T7A07. What is the function of a transceiver's PTT input?
A. Input for a key used to send CW
B. Switches transceiver from receive to transmit when grounded
C. Provides a transmit tuning tone when grounded
D. Input for a preamplifier tuning tone

T7A08. Which of the following describes combining speech with an RF carrier signal?
A. Impedance matching
B. Oscillation
C. Modulation
D. Low-pass filtering

T7A09. What is the function of the switch which selects either SSB or CW-FM on some VHF power amplifiers?
A. Change the mode of the transmitted signal
B. Set the amplifier for proper operation in the selected mode
C. Change the frequency range of the amplifier to operate in the proper segment of the band
D. Reduce the received signal noise

T7A10. What can be added to the output of a transceiver to increase the transmitted output power?
A. A potentiometer
B. An RF power amplifier
C. An impedance multiplier
D. All these choices are correct

T7A11. What is the function of the Variable Frequency Oscillator (VFO) circuit in a transceiver?
A. Set the receive and transmit frequency
B. Provide automatic frequency control
C. Inject a variable frequency to allow CW reception
D. Generate and demodulate single sideband signals

Answer Key

T7A01: B T7A02: A T7A03: B T7A04: C

T7A05: D T7A06: C T7A07: B T7A08: C

T7A09: B T7A10: B T7A11: A